

Tariq Khalaf

TMKhalaf0@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

University of North Carolina at Charlotte

Expected: Dec 2028

Bachelors of Science in Electrical and Computer Engineering

Skills

Design & Simulation: KiCad, Fusion 360, AutoCAD, SolidWorks, PCB Design, Circuit Diagrams, VS Code, Dev-C++, Meshmixer, 3D Printing.

Programming: Python, C++, JavaScript, Arduino, Embedded C, UART.

Engineering Skills: Soldering, PCB bring-up/debugging, USB-C power/programming, brushed DC and servo motor control, custom boards, circuit prototyping.

Experience

Undergraduate Research Assistant

Aug 2026 – Present

University of North Carolina at Charlotte

- Conducting research on SLAM (Simultaneous Localization and Mapping) for indoor positioning applications.
- Exploring RSSI-based signal analysis and trilateration techniques to estimate device location; project is ongoing.

Teacher's Assistant

Jan 2026 – Jul 2026

Central Piedmont Community College

- Assisted students during labs across 15+ projects, teaching hands-on techniques, equipment navigation, and troubleshooting builds.
- Supported hands-on work in electronics, coding, and mechanical/civil engineering builds using Arduino, circuits, sensors, and prototypes.

Projects

Brushed DC Electronic Speed Controller – Academic Project

Dec 2025 – Jan 2026

- Designed a 24V DC motor ESC supporting up to 5A continuous current, emphasizing safe power delivery and reliability.
- Created multi-layer PCB layouts in KiCad with ground planes, power pours, via stitching, and high-current trace planning.
- Selected parts using datasheets and supplier availability to balance performance, cost, footprint, and manufacturability.

Embedded MCU Power & Programming Board – Personal Project

Jun 2026 – Jul 2026

- Designed a custom MCU board accepting 24–36V input, stepped down to 3.3V using an onboard buck regulator.
- Integrated USB-C for board power/programming with required power, programming, and protection support circuitry.
- Planned PCB layout around regulator placement, decoupling, grounding, and clean 3.3V delivery for embedded hardware.

Automated Drawbridge on a Custom MCU Board – Academic Project

Apr 2026 – May 2026

- Built a motorized bridge that opens and closes automatically, running off a custom MCU board driving one brushed DC motor and a servo through a TB6612FNG dual H-bridge driver.
- Automated crossing logic from ultrasonic sensor input, with traffic LEDs and buzzers signaling bridge state to vehicles and boats.
- Fabricated the structure from a self-designed 3D-printed bridge deck on hand-cast cement pillars, integrating mechanics, power distribution, and control into one working prototype.

Remote-Control Robot – Personal Project

Jan 2026 – Feb 2026

- Built an ESP32 robot driving two brushed DC motors through a TB6612FNG dual H-bridge with wireless remote input.
- Integrated an ultrasonic sensor, motion detector, buzzers, and an onboard camera for obstacle detection and alerts.
- Wired power distribution and common grounds across driver, MCU, and sensors; tuned PWM drive logic for stability under load.

Personal A.I. Assistant – Personal Project

Jul 2026 – Aug 2026

- Built a hands-free voice assistant in Python running entirely on local hardware, pairing a speech-to-text engine with GPU neural text-to-speech for ~1–2s spoken replies.
- Implemented word activation, mid-reply interruption, and an intent router mapping speech to calendar, browser, and file actions.
- Added a real-time browser visualizer over WebSockets, streaming live audio and assistant state to an animated interface.

Door Alarm System – Personal Project

Jan 2026 – Feb 2026

- Built an ESP32-based door alarm with a reed switch, DFPlayer Mini, microSD card, and speaker to play custom audio alerts.
- Implemented UART communication and sensor-state logic to trigger MP3 playback when door/latch movement was detected.
- Designed and 3D-printed a custom enclosure sized to snugly fit each component, powered by a 4xAA battery pack for standalone operation.

Involvements

Charlotte Rocketry Club – UNC Charlotte

2026 – Present

- New member contributing to PCB design for club projects as the semester's build cycle gets underway.